

torch.nn.utils.rnn.pack_padded_sequence

https://pytorch.org/docs/stable/generated/torch.nn.utils.rnn.pack_padded_sequence

Description:

Packs a Tensor containing padded sequences of variable length. input can be of size $T \times B \times *$ (if batch_first is False) or $B \times T \times *$ (if batch_first is True) where T is the length of the longest sequence, B is the batch size, and $*$ is any number of dimensions (including 0). For unsorted sequences, use `enforce_sorted = False`. If `enforce_sorted` is True, the sequences should be sorted by length in a decreasing order, i.e. `input[:,0]` should be the longest sequence, and `input[:,B-1]` the shortest one. `enforce_sorted = True` is only necessary for ONNX export. It is an inverse operation to `pad_packed_sequence()`, and hence `pad_packed_sequence()` can be used to recover the underlying tensor packed in `PackedSequence`.

Function Signature

```
torch.nn.utils.rnn.pack_padded_sequence(input, lengths,  
batch_first=False, enforce_sorted=True)[source]#
```

Parameters

Return type

Returns

Return type

Returns:

PackedSequence

Notes & Warnings

NOTE

Note This function accepts any input that has at least two dimensions. You can apply it to pack the labels, and use the output of the RNN with them to compute the loss directly. A Tensor can be retrieved from a PackedSequence object by accessing its `.data` attribute.

⚠ WARNING

Warning The dim of input tensor will be truncated if its length larger than correspond value in length.

Detailed Documentation

Documentation

Packs a Tensor containing padded sequences of variable length.

input can be of size $T \times B \times *$ (if `batch_first` is `False`) or $B \times T \times *$ (if `batch_first` is `True`) where T is the length of the longest sequence, B is the batch size, and $*$ is any number of dimensions (including 0).

For unsorted sequences, use `enforce_sorted = False`. If `enforce_sorted` is `True`, the sequences should be sorted by length in a decreasing order, i.e. `input[:,0]` should be the longest sequence, and `input[:,B-1]` the shortest one. `enforce_sorted = True` is only necessary for ONNX export.

It is an inverse operation to `pad_packed_sequence()`, and hence `pad_packed_sequence()` can be used to recover the underlying tensor packed in `PackedSequence`.

This function accepts any input that has at least two dimensions. You can apply it to pack the labels, and use the output of the RNN with them to compute the loss directly. A Tensor can be retrieved from a `PackedSequence` object by accessing its `.data` attribute.

input (Tensor) – padded batch of variable length sequences.

lengths (Tensor or list(int)) – list of sequence lengths of each batch element (must be on the CPU if provided as a tensor).

batch_first (bool, optional) – if `True`, the input is expected in $B \times T \times *$ format, $T \times B \times *$

otherwise. Default: False.

`enforce_sorted` (bool, optional) – if True, the input is expected to contain sequences sorted by length in a decreasing order. If False, the input will get sorted unconditionally. Default: True.

The dim of input tensor will be truncated if its length larger than correspond value in length.

a `PackedSequence` object